

1 Lie Derivative of Forms and Derivative of Integrals

1.1 Interior Product (Contraction)

Definition 1.1 (Interior Product).

On a smooth manifold $\mathcal{M}$ of dimension m , let $v \in T_p\mathcal{M}$, $\alpha \in \Lambda^k(\mathcal{M})$. Then we define the interior product of the vector and the form to be

$$v \lrcorner \alpha = \begin{cases} 0 & \forall \alpha \in \Lambda^0(\mathcal{M}) \\ \alpha(v) & \forall \alpha \in \Lambda^1(\mathcal{M}) \\ \alpha(v, \dots) & \forall \alpha \in \Lambda^k(\mathcal{M}). \end{cases}$$

Theorem 1.1 (Basic Properties of Interior Product).

$$\begin{aligned} (v + cw) \lrcorner \alpha &= v \lrcorner \alpha + c(w \lrcorner \alpha) \\ v \lrcorner (\alpha + f\beta) &= v \lrcorner \alpha + f \cdot (v \lrcorner \beta) \end{aligned}$$

Theorem 1.2 (Interior Product and Cofactor Expansion).

For any $v \in T_p\mathcal{M}$ and k covectors $\omega^i \in \Lambda^1(\mathcal{M})$, $1 \leq i \leq k$, we have

$$v \lrcorner (\omega^1 \wedge \dots \wedge \omega^k) = \sum_{i=1}^k (-1)^{i-1} \omega^i(v) \omega^1 \wedge \dots \wedge \check{\omega^i} \wedge \dots \wedge \omega^k.$$

Proof. Form the matrix $\xi^i_j = \omega^i(v_j)$ as in ???. We see

$$(\omega^1 \wedge \dots \wedge \omega^k)(v_1, \dots, v_k) = \det(\xi^i_j).$$

Cofactor expand this on the first column, and the theorem is proved. $\square$

Corollary 1.2.1 (Interior Product as an Antiderivation).

For any $v \in T_p\mathcal{M}$, $\alpha \in \Lambda^{n_1}(\mathcal{M})$, $\beta \in \Lambda^{n_2}(\mathcal{M})$, we have

$$v \lrcorner (\alpha \wedge \beta) = (v \lrcorner \alpha) \wedge \beta + (-1)^{n_1} \alpha \wedge (v \lrcorner \beta).$$

Proof. Let $\alpha = \alpha_{i_1 \dots i_{n_1}} dx^{i_1} \wedge \dots \wedge dx^{i_{n_1}}$, $\beta = \beta_{j_1 \dots j_{n_2}} dx^{j_1} \wedge \dots \wedge dx^{j_{n_2}}$. Then

$$\begin{aligned} v \lrcorner (\alpha \wedge \beta) &= v \lrcorner (\alpha_{i_1 \dots i_{n_1}} \beta_{j_1 \dots j_{n_2}} dx^{i_1} \wedge \dots \wedge dx^{i_{n_1}} \wedge dx^{j_1} \wedge \dots \wedge dx^{j_{n_2}}) \\ &= \alpha_{i_1 \dots i_{n_1}} \beta_{j_1 \dots j_{n_2}} (v \lrcorner (dx^{i_1} \wedge \dots \wedge dx^{i_{n_1}} \wedge dx^{j_1} \wedge \dots \wedge dx^{j_{n_2}})). \end{aligned}$$

But

$$\begin{aligned} (v \lrcorner \alpha) \wedge \beta &= \alpha_{i_1 \dots i_{n_1}} \beta_{j_1 \dots j_{n_2}} v \lrcorner (dx^{i_1} \wedge \dots \wedge dx^{i_{n_1}}) \wedge dx^{j_1} \wedge \dots \wedge dx^{j_{n_2}} \\ \alpha \wedge (v \lrcorner \beta) &= \alpha_{i_1 \dots i_{n_1}} \beta_{j_1 \dots j_{n_2}} dx^{i_1} \wedge \dots \wedge dx^{i_{n_1}} \wedge v \lrcorner (dx^{j_1} \wedge \dots \wedge dx^{j_{n_2}}). \end{aligned}$$

If we expand all these using **Theorem 1.2**, we see that we need $\alpha \wedge (v \lrcorner \beta)$ to have a factor of $(-1)^{n_1}$, since we are essentially piecing two cofactors together. $\square$

Corollary 1.2.2 (Coordinate Representation).

$$(v \lrcorner \alpha)_{i_1 \dots i_k} = v^{i_0} \alpha_{i_0 i_1 \dots i_k}.$$

1.2 Lie Derivative of Forms

Definition 1.2 (Lie Derivative of Forms).

Given a form $\omega \in \Lambda^k(\mathcal{M})$ and a vector field X , we define the Lie derivative of ω by X to be a k -form,

$$(\mathcal{L}_X \omega)_p = \frac{d}{dt} (\sigma_t^{X*} \omega)_p \Big|_{t=0} = \lim_{t \rightarrow 0} \frac{\sigma_t^{X*} (\omega_{\sigma_t^X(p)}) - \omega_p}{t}$$

Theorem 1.3 (Lie Derivative of Functions).

This is essentially a rephrase of ??, but this time we see f as a form.

$$\mathcal{L}_X f = Xf.$$

Proof.

$$\mathcal{L}_X f = \frac{d}{dt} (\sigma_t^{X*} f) \Big|_{t=0} = \frac{d}{dt} (f \circ \sigma_t^X) \Big|_{t=0} = Xf.$$

$\square$

Theorem 1.4 (Various Product Rules).

$$\begin{aligned}\mathcal{L}_X(f\omega) &= (\mathcal{L}_X f)\omega + f\mathcal{L}_X\omega \\ \mathcal{L}_X(\alpha \otimes \beta) &= (\mathcal{L}_X\alpha) \otimes \beta + \alpha \otimes (\mathcal{L}_X\beta).\end{aligned}$$

Corollary 1.4.1.

$$\mathcal{L}_X(\alpha \wedge \beta) = (\mathcal{L}_X\alpha) \wedge \beta + \alpha \wedge (\mathcal{L}_X\beta).$$

Theorem 1.5 (Lie Derivative of Forms Paired with Vector Fields).

$$\begin{aligned}\mathcal{L}_X(\omega(X_1, \dots, X_k)) \\ = (\mathcal{L}_X\omega)(X_1, \dots, X_k) + \omega(\mathcal{L}_X X_1, \dots, X_k) + \dots + \omega(X_1, \dots, \mathcal{L}_X X_k).\end{aligned}$$

Corollary 1.5.1 (Lie Derivative of Forms as Directional Derivatives and Commutators).

$$\begin{aligned}(\mathcal{L}_X\omega)(X_1, \dots, X_k) \\ = X(\omega(X_1, \dots, X_k)) - \omega([X, X_1], \dots, X_k) - \dots - \omega(X_1, \dots, [X, X_k]).\end{aligned}$$

Theorem 1.6 (Cartan's Magic Formula).

$$\mathcal{L}_\omega = X \lrcorner (d\omega) + d(X \lrcorner \omega).$$

Corollary 1.6.1 (Lie Derivative Commute with Exterior Derivative).

$$\mathcal{L}_X(d\omega) = d(\mathcal{L}_X\omega).$$